Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Releaf

Project / Solution Name

Releaf

Name (Leader)

Lin Jiarun

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Live Prototype URL

https://releaf-prometheusjens.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/sjq98/Releaf

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The application is one Node.js 22 process serving a browser UI and a JSON API over the same origin.

The browser loads static ES modules; every business call carries a seven-day HttpOnly session cookie. The server owns five concerns: local accounts with administrator and member roles, a shared store for tables, records, references, versions and field history, a fixed warehouse service that turns inbound and outbound documents into approved, posted stock movements, an independent order space, and a meeting-summary importer. Totals, line states and balances are computed in one module called by both the UI routes and the Agent tools, so no caller can define a different rule.

All persistent state — business tables, the immutable movement ledger, sessions, assistant conversations and audit entries — lives in one SQLite file through Node's built-in `node:sqlite`.

One integration point leaves the process: outbound HTTPS to an OpenAI-compatible Chat Completions endpoint, used for order intake, meeting-to-task extraction and assistant tool calling, constrained by strict schemas and validated server-side before any record is written. Deployment is currently local loopback.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js 22.13+; SQLite / node:sqlite	HTTP runtime, tests and local persistence.	Node: MIT + bundled notices; SQLite: public domain
HTML / CSS / ES modules; Lucide 1.60.0	Team-written browser UI; third-party interface icons.	UI: project Apache-2.0; Lucide: ISC + MIT
ExcelJS 4.4.0; csv-parse 7.0.3	XLSX / UTF-8 CSV parsing, field mapping and import.	MIT
pdfjs-dist 6.4.299	Browser PDF attachment rendering.	Apache-2.0
decimal.js 10.6.0; jsep 1.4.0	Decimal quantity arithmetic and restricted formula parsing.	MIT
Chart.js 4.6.1	Locally served inventory dashboard charts.	MIT

Optional separate bridge: whatsapp-digest-local (MIT), whatsapp-web.js and qrcode; see THIRD_PARTY_NOTICES.md. Not bundled.

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek Chat Completions API	Order and meeting-task extraction; Assistant tool calling; model discovery.	Paid, usage-based API access
WhatsApp Web (Meta) via local whatsapp-web.js bridge	Optional selected-chat order intake via a local bridge. No official WhatsApp Cloud API, outbound messaging or payment verification. Live account integration not verified.	N/A - local bridge; no paid Cloud API used
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A

API reference: api-docs.deepseek.com (checked 25 Oct 2026).
Releaf's own Agent APIs are product interfaces, not third-party APIs.

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development planning and team collaboration.	Core (team-reported); region not selected
Local Node.js / SQLite	Current reproducible runtime, storage and local account authentication.	N/A - local machine
Replit App hosting	Team-supplied HTTPS URL in section 1. Page reachable; authentication endpoint returned 403 in the latest check.	Plan / region not verified
RunPod	Not used. No team-hosted model training, fine-tuning or GPU runtime. Hosted model access is through DeepSeek.	N/A - no Pod or endpoint
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
deepseek-flash (API model ID)	DeepSeek API	Pre-trained; hosted inference. No team fine-tuning.	DeepSeek Terms of Use and Privacy Policy
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

Version note (25 Oct 2026): deepseek-flash is a provider-managed alias.
DeepSeek currently serves DeepSeek-V4.1-Flash-1017 for this model ID.
References: DeepSeek Models & Pricing; DeepSeek Terms of Use.

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Inventory demo	Team-created scripts	Synthetic	Team-owned	N
Meeting / order examples	Team-created examples	Synthetic	Team-owned	N
Tests / mocked AI	Repository tests/	Synthetic	Project Apache-2.0	N
N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT / Codex 6.1-sol (team-reported)	Code, refactoring, tests and documentation.	Major
Claude Opus / Sonnet 5.5 (team-reported)	Architecture review and submission document drafting.	Moderate
N/A	N/A	N/A
N/A	N/A	N/A

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team retains responsibility for committed code and submitted materials. Automated tests cover validation, account and Agent permissions, warehouse posting and reversal, orders, spreadsheet imports and server behaviour. Provider responses are mocked; these tests do not establish live-model accuracy. Business AI outputs are schema-validated. Assistant writes default to Review first; explicit Full access still enforces permissions and workflow rules. Meeting extraction creates tasks for subsequent human review. AI output does not establish payment receipt or authorize stock posting.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Platform / development Agent	Agent-assisted planning and team collaboration.
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A

Evidence appendix: the team's supplied Replit usage and project screenshots follow the signed declaration. Live hosting URL: https://releaf-prometheusjens.replit.app

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supported development planning, Agent-assisted work and collaboration; it is not a required dependency of the core business logic and is replaceable with local Node.js and GitHub. The Replit App page is reachable; authentication configuration still requires verification. The supplied proof records platform use, not a product-level Replit API integration. The hosted AI provider is configurable by endpoint, model and key. Manual warehouse validation, approval, posting, reversal, ledger and dashboard work without an AI provider. No RunPod runtime or credits were used.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo uses synthetic business examples. Accounts, display names, roles, records, attachments, source messages, meeting summaries, audit history and assistant conversations can be stored locally. No automatic retention policy is implemented. Passwords use salted scrypt hashes; session tokens are hashed. Sessions use seven-day HttpOnly, SameSite=Strict cookies. Provider keys stay server-side but are not encrypted at rest. Access follows account roles and Agent token scope.

Selected order text, meeting summaries and relevant assistant context may be transmitted to the configured AI provider over HTTPS. Optional WhatsApp access reads explicitly selected chats through a local bridge. Provider retention and training practices are governed by its actual terms and settings; this prototype does not guarantee no retention or no training. Production use requires consent, minimization, retention controls and an appropriate provider agreement.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI can omit or misinterpret order details, task owners and deadlines. Structured validation does not guarantee semantic accuracy. Orders start unreviewed, with no AI-entered amount; payment and invoice records need human confirmation. Meeting tasks retain source quotes and require review. Orders do not reserve or move stock.

Stock arithmetic and warehouse approval/posting are deterministic and independent of the AI provider. Default Assistant write previews, role checks, versions and idempotency reduce accidental changes; explicit Full access is still subject to business rules.

Provider outages or model changes can interrupt AI intake. Configurable providers and manual workflows reduce dependence. The prototype uses a shared workspace, not tenant isolation. Public hosting verification, secure secret storage, backup/retention policies and pilot accuracy measurements remain outstanding.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

Status on 25 October 2026: Live URL supplied; the page is reachable, but authentication returned 403. Functional judge access and contact email consistency still require confirmation.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

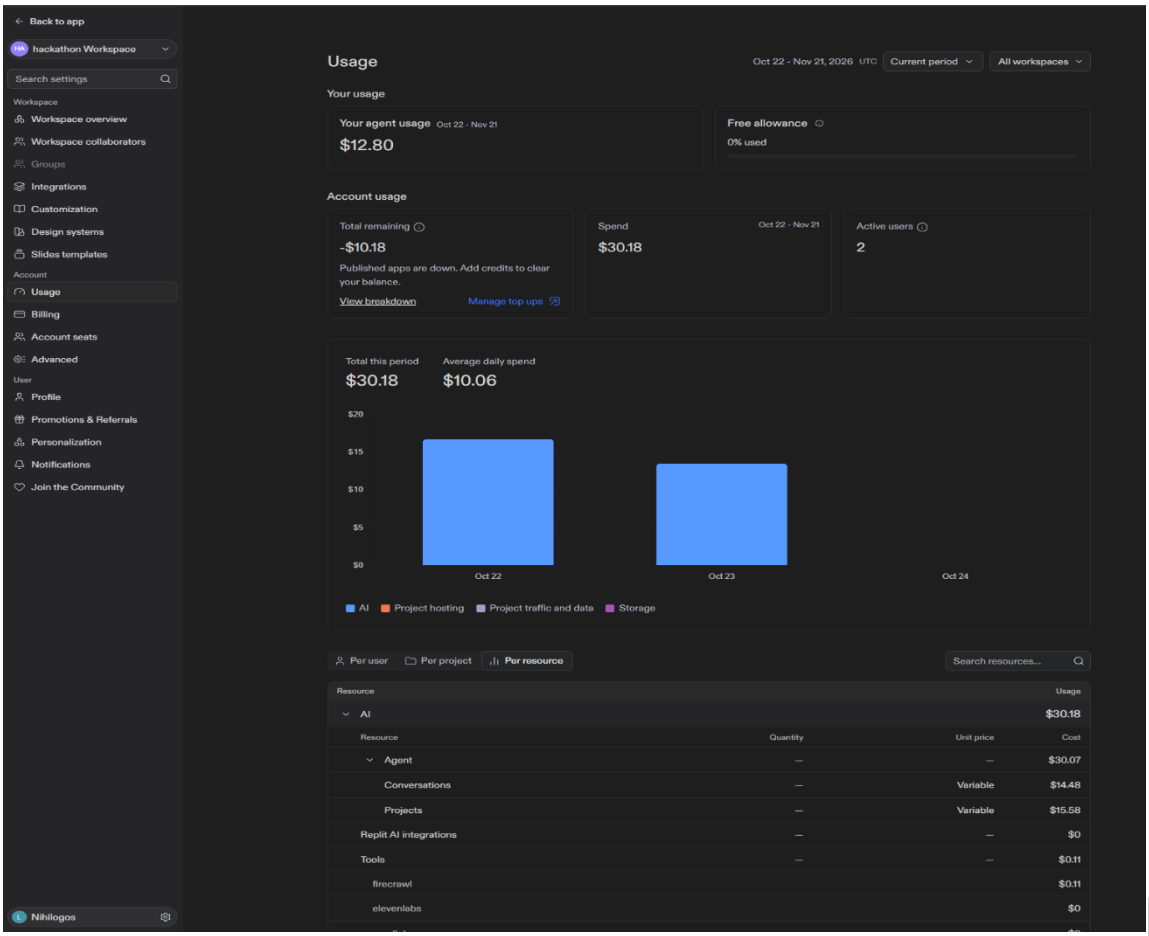
[Signature]

Lin Jiarun

[Date]

2026/12/25

Replit account: nihilogos@outlook.com



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